

Solution Architecture

Fit Buddy AI – AI Fitness Plan Generator using Gemini Models

Generative AI Project | FastAPI + Google Gemini

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Layers

- Presentation: HTML / CSS / Jinja2 templates
- Application: FastAPI routes and request handling
- Service: Gemini integration, fallback generator and serializers
- Data: SQLAlchemy models and SQLite database

Flow

Browser → FastAPI → Gemini service (or fallback) → structured 7-day plan → SQLite → Browser

Design principle

Keep UI, routing, services and data responsibilities separated so the project is easier to maintain and demonstrate. The architecture diagram is on the next page.

Architecture Diagram

